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IPv6静态路由实验

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IPv6地址

➤ IPv6地址由8位16进制组成

➤ IPv6地址简写方法有三种：

1， 每组16进制数前面的0可以省略不写

完整： 2403:A200:A200:1100:0000:0000:0F00:0003

简写： 2403:A200:A200:1100:0000:0000:F00:3

2， 连续为0的以组或多组可以用：：代替， 一个IP中只能用一次：：， 否则系统回识别错误

完整： 2403:A200:A200:1100:0000:0000:0F00:0003

简写： 2403:A200:A200:1100::F00:3

3， 如果一组中全是0， 又不想用：：代替， 每组中要保留1个0

完整： 2403:A200:A200:0000:AFFE:0000:0000:0003

简写： 2403:A200:A200:0:AFFE::3

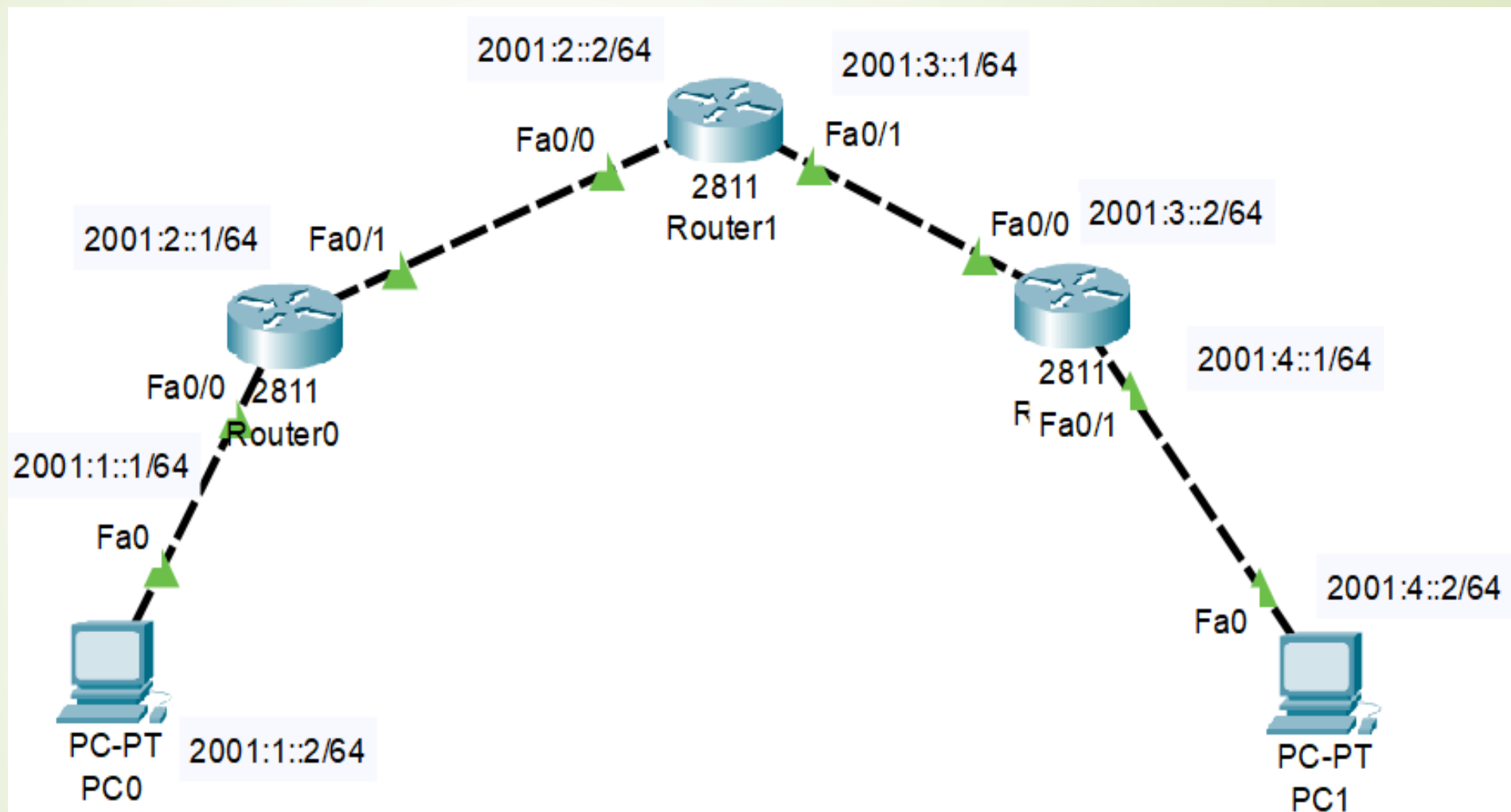
实验目的

- 熟悉Ipv6地址的表示方法
- 掌握Ipv6地址的配置方法

实验步骤

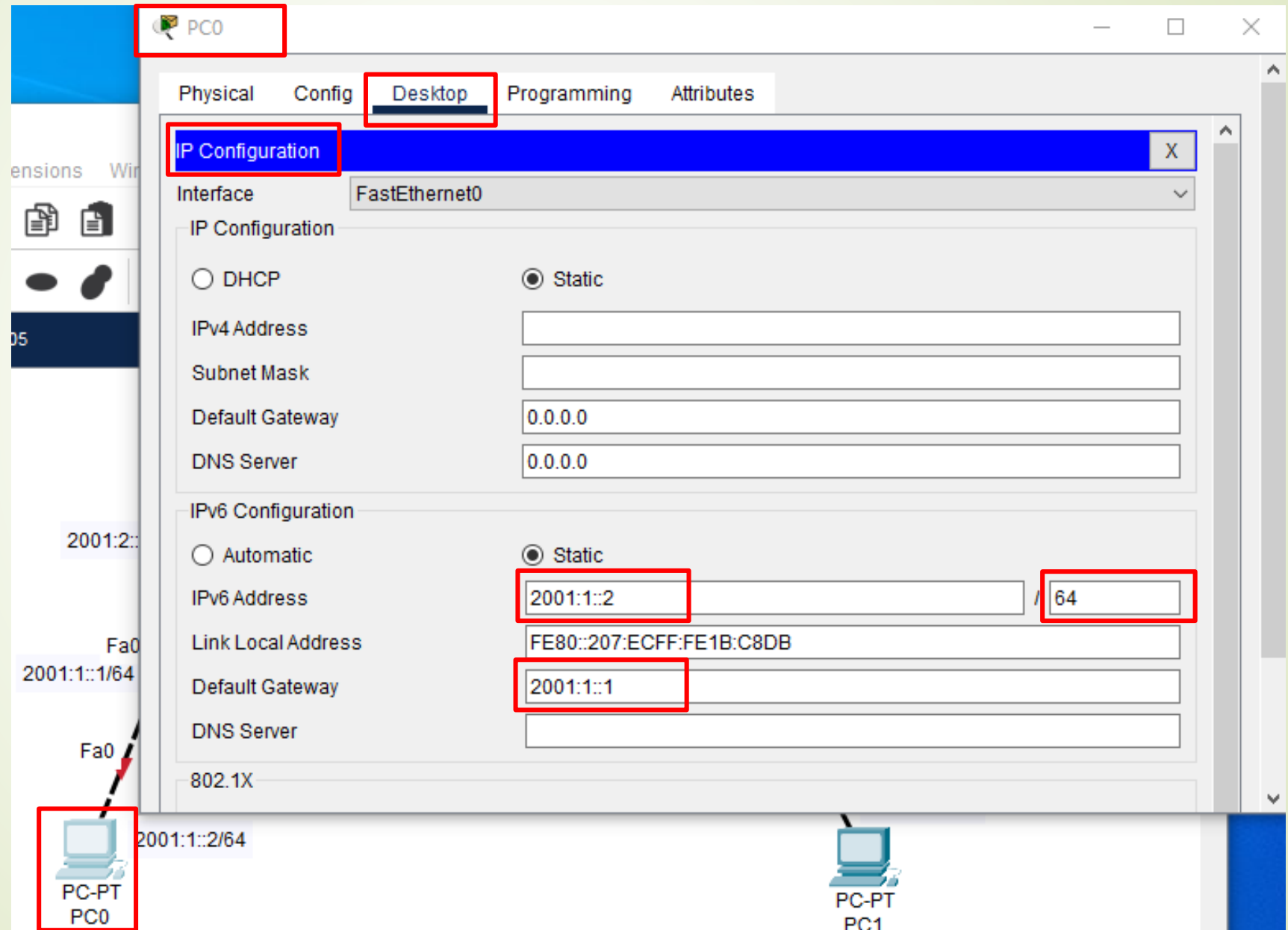
- ➡ 1, 规划网络地址及拓扑图;
- ➡ 2, 配置所有设备的IP、网关、子网掩码
- ➡ 3, 检查网络连通性
- ➡ 4, 配置静态路由
- ➡ 5, 再次检查网络连通性, 并查看路由表的信息变化

步骤一：按照拓扑图搭建网络结构

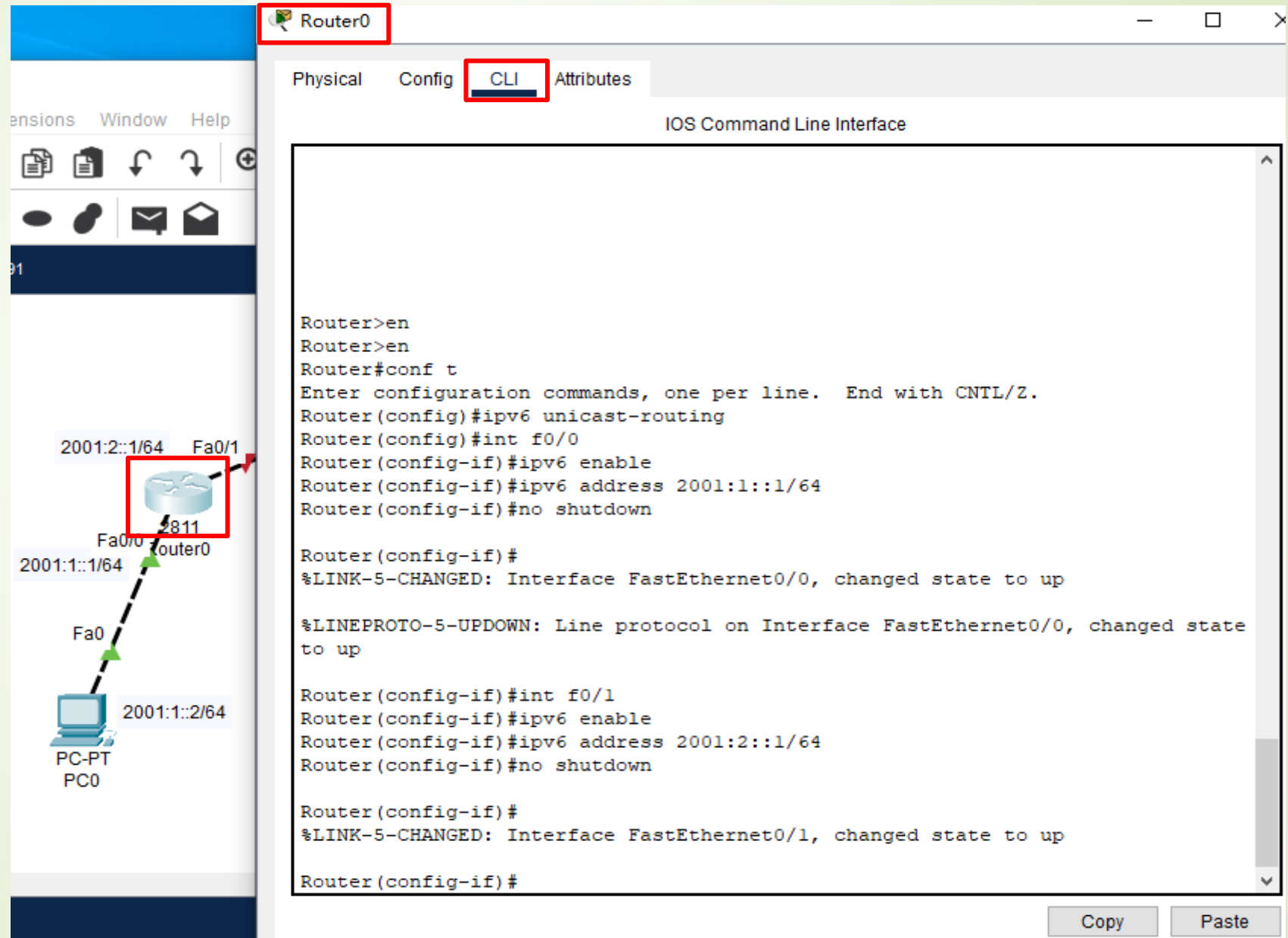


步骤二：配置PC机Ipv6地址（以PC0为例）

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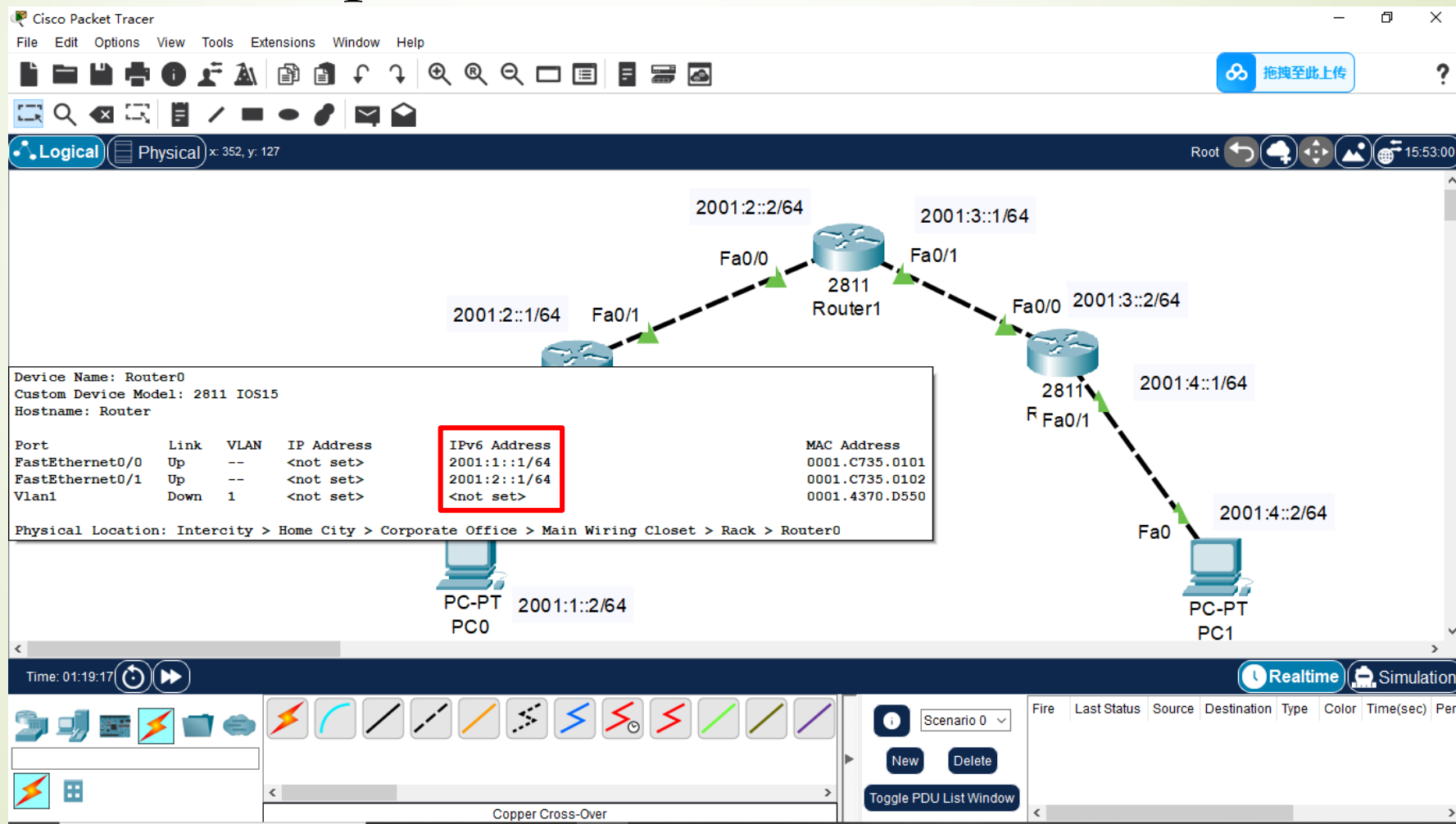


步骤二：各条命令说明

Enable	进入特权模式
conf term	进入全局配置模式
ipv6 unicast-routing	开启单播功能（这步先做，顺序不能错）
int f0/0	进入接口f0/0
ipv6 enable	开启ipv6功能
ipv6 address 2001:1::1/64	配置IPv6地址和子网掩码
no shutdown	启动端口
int f0/1	进入接口f0/1
ipv6 enable	开启ipv6功能
ipv6 address 2001:2::1/64	配置IPv6地址和子网掩码
no shutdown	启动端口

步骤三：检查

此时连线上的箭头都是绿色，并且各设备/端口都有Ipv6地址



步骤三：检查整个网络的连通性

PCO

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 2001:1::1

Pinging 2001:1::1 with 32 bytes of data:

Reply from 2001:1::1: bytes=32 time<1ms TTL=255
Reply from 2001:1::1: bytes=32 time<1ms TTL=255
Reply from 2001:1::1: bytes=32 time<1ms TTL=255
Reply from 2001:1::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:1::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:2::1

Pinging 2001:2::1 with 32 bytes of data:

Reply from 2001:2::1: bytes=32 time<1ms TTL=255
Reply from 2001:2::1: bytes=32 time<1ms TTL=255
Reply from 2001:2::1: bytes=32 time<1ms TTL=255
Reply from 2001:2::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:2::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

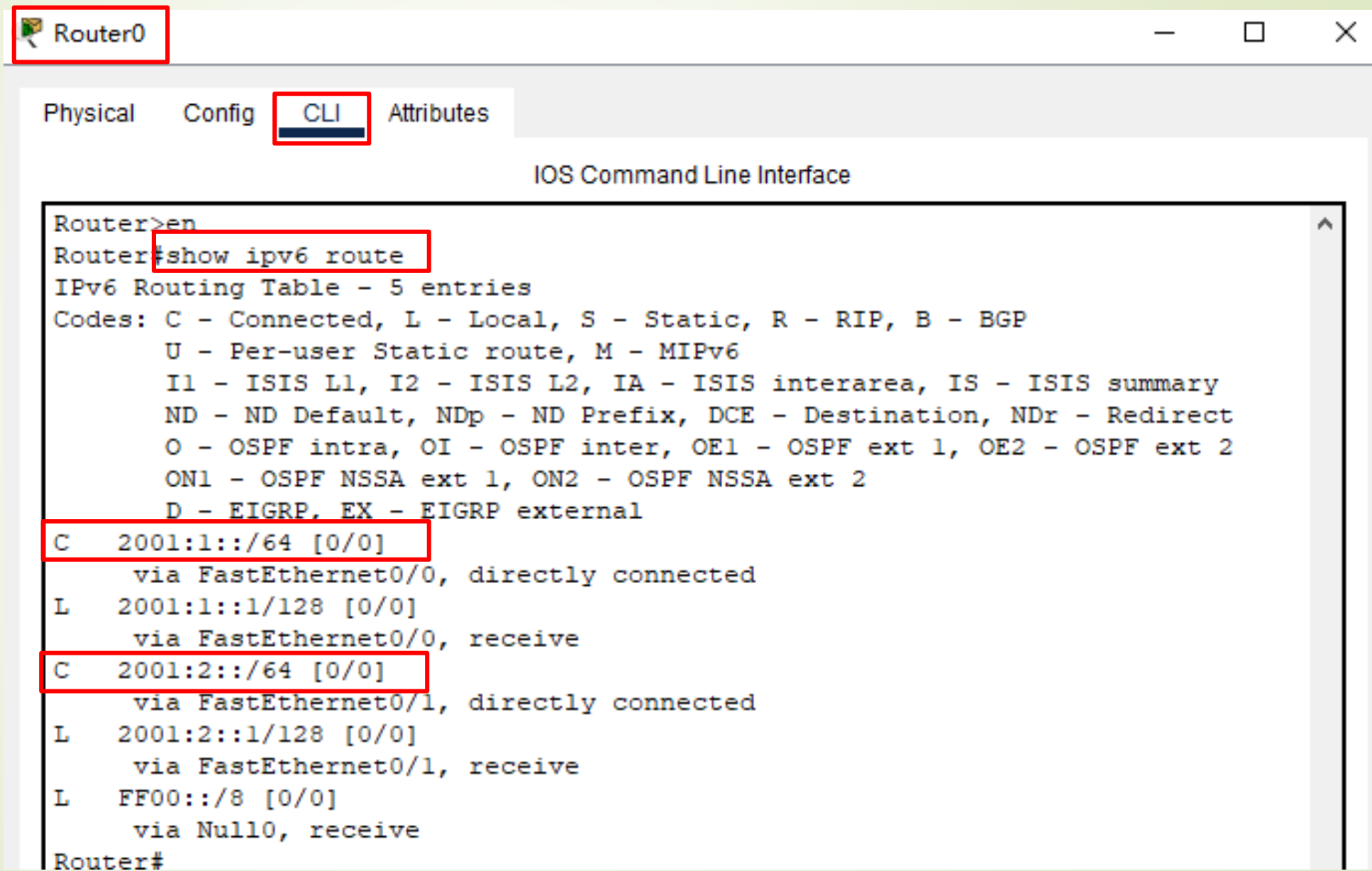
C:\>ping 2001:2::2

Pinging 2001:2::2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 2001:2::2:
```

步骤三：



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router>en
Router#show ipv6 route
IPv6 Routing Table - 5 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
C    2001:1::/64 [0/0]
      via FastEthernet0/0, directly connected
L    2001:1::1/128 [0/0]
      via FastEthernet0/0, receive
C    2001:2::/64 [0/0]
      via FastEthernet0/1, directly connected
L    2001:2::1/128 [0/0]
      via FastEthernet0/1, receive
L    FF00::/8 [0/0]
      via Null0, receive
Router#
```

步骤四：路由器添加路由(以Route0为例)

Router0

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
Router#conf term
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ipv6 route 2001:3::/64 2001:2::2
Router(config)#ipv6 route 2001:4::/64 2001:2::2
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

步骤五：再验证整个网络的连通性

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PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 2001:4::2
```

Pinging 2001:4::2 with 32 bytes of data:

```
Reply from 2001:4::2: bytes=32 time=1ms TTL=125
Reply from 2001:4::2: bytes=32 time=7ms TTL=125
Reply from 2001:4::2: bytes=32 time<1ms TTL=125
Reply from 2001:4::2: bytes=32 time<1ms TTL=125
```

Ping statistics for 2001:4::2:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 2ms
```

C:\>

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:1::1
```

Pinging 2001:1::1 with 32 bytes of data:

```
Reply from 2001:1::1: bytes=32 time<1ms TTL=253
Reply from 2001:1::1: bytes=32 time<1ms TTL=253
Reply from 2001:1::1: bytes=32 time<1ms TTL=253
Reply from 2001:1::1: bytes=32 time<1ms TTL=253
```

Ping statistics for 2001:1::1:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

C:\>

步骤五：查看路由器变化(以Route0为例)

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router#conf term
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ipv6 route 2001:3::/64 2001:2::2
Router(config)#ipv6 route 2001:4::/64 2001:2::2
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ipv6 route
IPv6 Routing Table - 7 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
C    2001:1::/64 [0/0]
     via FastEthernet0/0, directly connected
L    2001:1::1/128 [0/0]
     via FastEthernet0/0, receive
C    2001:2::/64 [0/0]
     via FastEthernet0/1, directly connected
L    2001:2::1/128 [0/0]
     via FastEthernet0/1, receive
S    2001:3::/64 [1/0]
     via 2001:2::2
S    2001:4::/64 [1/0]
     via 2001:2::2
L    FF00::/8 [0/0]
     via Null0, receive
Router#
```